Heart Stroke Prediction

Executed project report with saved notebook outputs, tables, and charts

Project Description

Heart Stroke Prediction

Project Summary

This project is a clean, student-friendly implementation of a stroke risk prediction. The main goal is to predict stroke risk from patient health and lifestyle attributes. The notebook keeps the workflow simple: load the dataset, understand the columns, clean the data, train a baseline model, and check how well the model performs.

No external repository links or profile links are included in this description. Everything needed to understand and run the project is kept inside this folder.

Problem Statement

A raw dataset is useful only when it is converted into a clear decision-making workflow. In this project, the data is processed step by step so that important patterns can be found and a machine learning model can be trained. The expected output is based on stroke.

Files Included

File	Purpose	Quick Note
healthcare-dataset-stroke-data.csv	Main data source for this folder	2000 sampled rows x 12 columns
Notebook file	Complete Python workflow	Includes loading, cleaning, training, and evaluation
PDF report	Human-readable project summary	Matches the updated project structure

Important Columns

Column	Role
id	Input feature used in the modelling workflow
gender	Input feature used in the modelling workflow
age	Input feature used in the modelling workflow
hypertension	Input feature used in the modelling workflow
heartdisease	Input feature used in the modelling workflow
evermarried	Input feature used in the modelling workflow
worktype	Input feature used in the modelling workflow
Residencetype	Input feature used in the modelling workflow
avgglucoselevel	Input feature used in the modelling workflow
bmi	Input feature used in the modelling workflow
smokingstatus	Input feature used in the modelling workflow
stroke	Target / output column used in the modelling workflow

Workflow Followed

1. Load the data from the local dataset file present in the same project folder.
2. Inspect the structure using shape, column names, missing values, and basic statistics.
3. Clean the dataset by removing empty columns, handling missing values, and keeping only useful features.
4. Prepare the model input by separating features and the target column.
5. Train a baseline model using a simple scikit-learn pipeline with preprocessing.
6. Evaluate the result using practical metrics such as accuracy, classification report, MAE, RMSE, or R2 score depending on the target type.
7. Write observations in a short and direct way so the project can be explained easily.

How to Run

1. Open the notebook in Jupyter Notebook, JupyterLab, Google Colab, or VS Code.
2. Keep the dataset files in the same folder as the notebook.
3. Run the cells from top to bottom.
4. Read the printed metrics and notes at the end before making conclusions.

Final Note

This version keeps the logic understandable and avoids unnecessary complexity. The aim is not to make the notebook look overloaded, but to make it readable, explainable, and useful for a student-level data science portfolio.

Executed Notebook Outputs

The outputs below were generated from the saved notebook after running the code cells in this project folder.

Cell 2 - 2. Load the dataset

```
Dataset shape: (5110, 12)
```

```
id gender age hypertension heart_disease ever_married \
0 9046 Male 68.6349 0 1 Yes
1 51676 Female 60.8834 0 0 Yes
2 31112 Male 81.7966 0 1 Yes
3 60182 Female 50.2352 0 0 Yes
4 1665 Female 78.9482 1 0 Yes

work_type Residence_type avg_glucose_level bmi smoking_status \
0 Private Urban 228.6165 37.5060 formerly smoked
1 Self-employed Rural 199.9754 NaN never smoked
2 Private Rural 106.5454 32.8696 never smoked
3 Private Urban 175.2145 35.2014 smokes
4 Self-employed Rural 174.0654 24.4872 never smoked

stroke
0 1
1 1
2 1
3 1
4 1
```

Cell 3 - 3. Understand the data

```
Rows: 5110
Columns: 12
```

```
column dtype missing_values unique_values
0 id int64 0 5110
1 gender object 0 3
2 age float64 0 5087
3 hypertension int64 0 2
4 heart_disease int64 0 2
5 ever_married object 0 2
6 work_type object 0 5
7 Residence_type object 0 2
8 avg_glucose_level float64 0 5099
9 bmi float64 201 4868
10 smoking_status object 0 4
11 stroke int64 0 2
```

Cell 4 - 4. Clean the data

```
Shape after simple cleaning: (5110, 12)
```

```
id gender age hypertension heart_disease ever_married \
0 9046 Male 68.6349 0 1 Yes
1 51676 Female 60.8834 0 0 Yes
2 31112 Male 81.7966 0 1 Yes
3 60182 Female 50.2352 0 0 Yes
4 1665 Female 78.9482 1 0 Yes

work_type Residence_type avg_glucose_level bmi smoking_status \
0 Private Urban 228.6165 37.5060 formerly smoked
1 Self-employed Rural 199.9754 NaN never smoked
2 Private Rural 106.5454 32.8696 never smoked
3 Private Urban 175.2145 35.2014 smokes
4 Self-employed Rural 174.0654 24.4872 never smoked

stroke
0 1
1 1
2 1
3 1
4 1
```

Cell 5 - 5. Select the target column

```
Selected target column: stroke
```

```
0 1
1 1
2 1
3 1
4 1
Name: stroke, dtype: int64
```

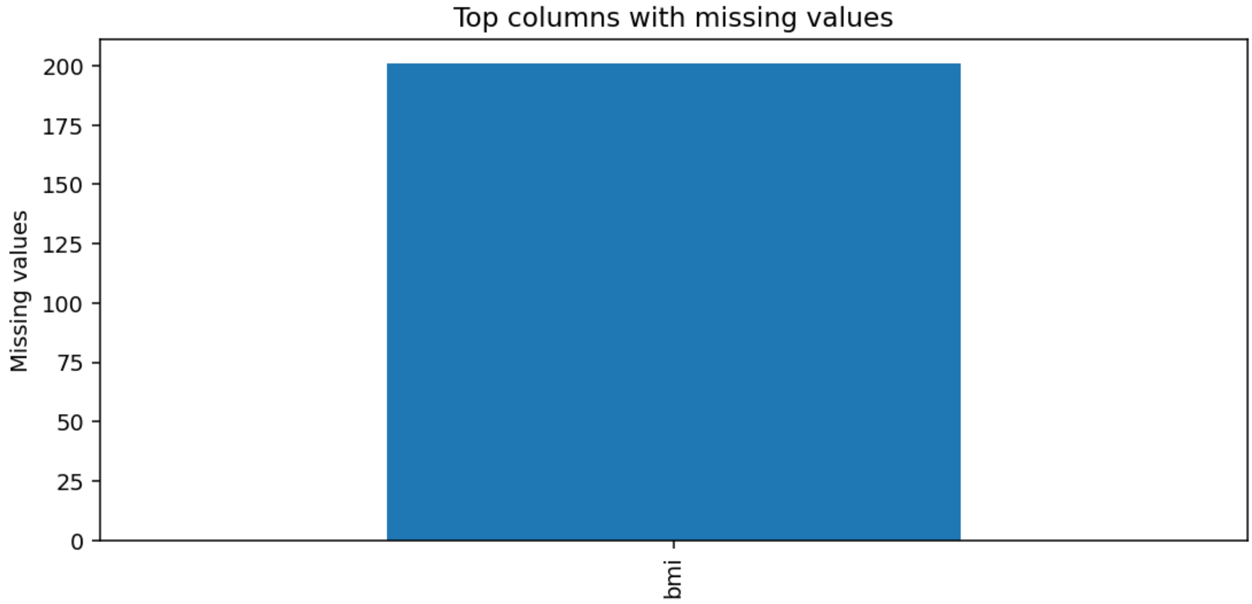
Cell 6 - 6. Quick EDA

```
Numeric columns: 7
Categorical columns: 5
```

```
count mean std min 25% \
id 5110.0 36517.829354 21161.721625 67.0000 17741.250000
age 5110.0 43.233953 22.633479 0.0799 25.287075
hypertension 5110.0 0.097456 0.296607 0.0000 0.000000
heart_disease 5110.0 0.054012 0.226063 0.0000 0.000000
avg_glucose_level 5110.0 106.153988 45.310994 53.3520 76.995500
```

```
bmi 4909.0 28.894147 7.856888 10.5535 23.588400
stroke 5110.0 0.048728 0.215320 0.0000 0.000000

50% 75% max
id 36932.0000 54682.00000 72940.0000
age 44.6798 60.88535 84.7926
hypertension 0.0000 0.00000 1.0000
heart_disease 0.0000 0.00000 1.0000
avg_glucose_level 92.0141 114.32660 271.3583
bmi 28.0590 33.11900 94.3910
stroke 0.0000 0.00000 1.0000
```



Cell 7 - 7. Prepare features and target

```
Dropped ID-like columns: ['id', 'age', 'Residence_type', 'avg_glucose_level']
Using 1200 sampled rows for model training speed.
Problem type: classification
Feature shape: (1200, 7)
Target unique values: 2
```

Cell 8 - 8. Train a baseline model

```
Training completed.
```

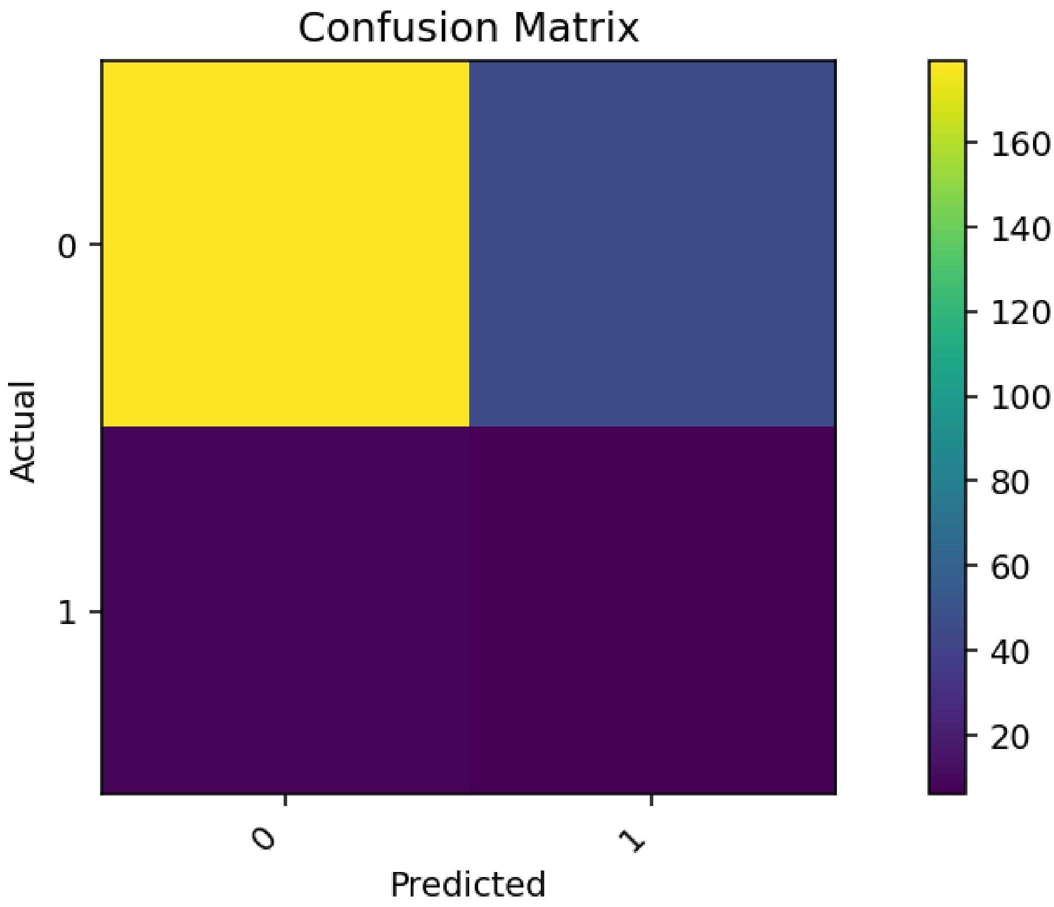
Cell 9 - 9. Evaluate the model

```
Accuracy: 0.7708

Classification report:
precision recall f1-score support

0 0.95 0.80 0.87 225
1 0.12 0.40 0.18 15

accuracy 0.77 240
macro avg 0.53 0.60 0.52 240
weighted avg 0.90 0.77 0.82 240
```


Confusion Matrix
Actual
0
1
Predicted
0
1
160
140
120
100
80
60
40
20